

Solardach Energie Dashboard: IEEE-Style Project Report on an Interactive PV-BESS Simulation and Tariff Analytics Tool for German Residential Energy Systems

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Abstract - This project presents the Solardach Energie Dashboard, an interactive Python-based tool for simulating residential photovoltaic-battery energy storage system (PV-BESS) performance across selected German cities. The dashboard integrates real hourly solar irradiance data acquisition from the NASA POWER API, synthetic but seasonally and diurnally shaped household demand modelling, hourly PV generation estimation, battery state-of-charge simulation, electricity tariff comparison, annual savings evaluation, payback estimation and CO2 avoidance calculation. The implementation is designed for Google Colab and uses pandas, NumPy, Plotly, Matplotlib and ipywidgets to create a video-ready dashboard and export CSV, HTML, PNG and ZIP outputs. The project is relevant to renewable energy management because it connects solar resource assessment, household load behaviour, storage dispatch, tariff sensitivity and energy independence into a single decision-support workflow.

Index Terms - PV-BESS, residential solar, battery storage, German tariffs, NASA POWER, self-sufficiency, payback period, smart grid analytics, Python dashboard.

I. INTRODUCTION

Residential solar PV combined with battery energy storage is becoming an important pathway for decentralised clean energy, household electricity cost reduction and grid flexibility. In the German context, techno-economic performance depends strongly on local solar resource, annual household demand, import tariff, feed-in revenue and battery capacity. The submitted code addresses this need through an interactive dashboard named Solardach Energie Dashboard, authored under the name Oihika Arpit. The tool allows users to select a German city, year, PV capacity, battery capacity, annual load, tariff case and performance ratio, and then observe the hourly and monthly energy behaviour of a PV-BESS system.

Unlike a static calculator, the project uses an hourly simulation logic. This is important because battery value is determined not only by annual PV generation but also by the timing mismatch between solar production and household consumption. The dashboard therefore models direct self-consumption, surplus charging, battery discharge during deficits, grid import, grid export and state of charge at each time step.

II. PROJECT OBJECTIVES

- 1) To build an interactive PV-BESS simulation dashboard for selected German cities.
- 2) To use hourly solar irradiance data and convert it into PV generation using a configurable performance ratio.
- 3) To create a realistic hourly residential demand profile using morning/evening peaks, seasonal effects and weekend effects.
- 4) To simulate battery charging and discharging with round-trip efficiency and C-rate limits.
- 5) To compare electricity bill savings under multiple German tariff assumptions.
- 6) To generate exportable CSV, HTML, PNG and ZIP outputs for reporting and video presentation.

III. INPUT DATA AND SYSTEM PARAMETERS

The implementation defines nine German cities grouped into three tiers. The city inputs include latitude, longitude and tier classification. These geographical coordinates are used to request hourly all-sky surface shortwave downward irradiance from the NASA POWER hourly point API. The code also defines four tariff cases for import electricity price, feed-in price and standing charge. User-adjustable dashboard inputs include city, year, PV capacity, battery capacity, annual demand, tariff selection, performance ratio and a refresh option for solar data caching.

Table I. German city dataset used in the dashboard.

Tier	City	Latitude	Longitude
Tier 1	Berlin	52.5200	13.4050
Tier 1	Munich	48.1351	11.5820
Tier 1	Hamburg	53.5511	9.9937

Tier 2	Freiburg	47.9990	7.8421
Tier 2	Leipzig	51.3397	12.3731
Tier 2	Dresden	51.0504	13.7373
Tier 3	Oldenburg	53.1435	8.2146
Tier 3	Cottbus	51.7563	14.3329
Tier 3	Passau	48.5667	13.4319

Table II. Tariff assumptions embedded in the submitted code.

Tariff Case	Import Price EUR/kWh	Feed-in Price EUR/kWh	Standing Charge EUR/year
BDEW average household	0.3700	0.0778	130
Low-cost new contract	0.2683	0.0778	120
Basic supply / Grundversorgung	0.4283	0.0778	150
Conservative 10-40 kWp case	0.3400	0.0673	130

IV. METHODOLOGY

The project methodology follows a data-driven simulation pipeline. First, hourly irradiance data are fetched or read from a local cache. Second, a normalized hourly demand profile is created from Gaussian-like morning and evening peaks, adjusted for winter, shoulder season, summer and weekend effects. Third, solar irradiance is converted to PV energy output. Fourth, the PV-BESS dispatch routine allocates PV energy to direct load service, battery charging, grid export, battery discharge and remaining grid import. Fifth, economic and environmental indicators are computed.

$$E_{PV}(t) = G(t)/1000 \times P_{PV} \times PR$$

where $E_{PV}(t)$ is hourly PV generation in kWh, $G(t)$ is hourly solar irradiance in Wh/m², P_{PV} is installed PV capacity in kWp and PR is the PV performance ratio. In the dashboard, PR is user-adjustable between 0.70 and 0.92, with 0.82 as the default value.

$$Annual\ Savings = Baseline\ Bill - New\ Grid\ Bill + Feed-in\ Revenue$$

$$Payback\ Period = Total\ System\ Cost / Annual\ Savings$$

The cost model uses EUR 1300/kWp for PV capacity, EUR 650/kWh for battery capacity and EUR 1800 as an installation cost. The carbon metric uses 0.38 kg CO₂/kWh multiplied by annual self-consumed PV energy.

Fig. 1. System architecture of the Solardach Energie Dashboard

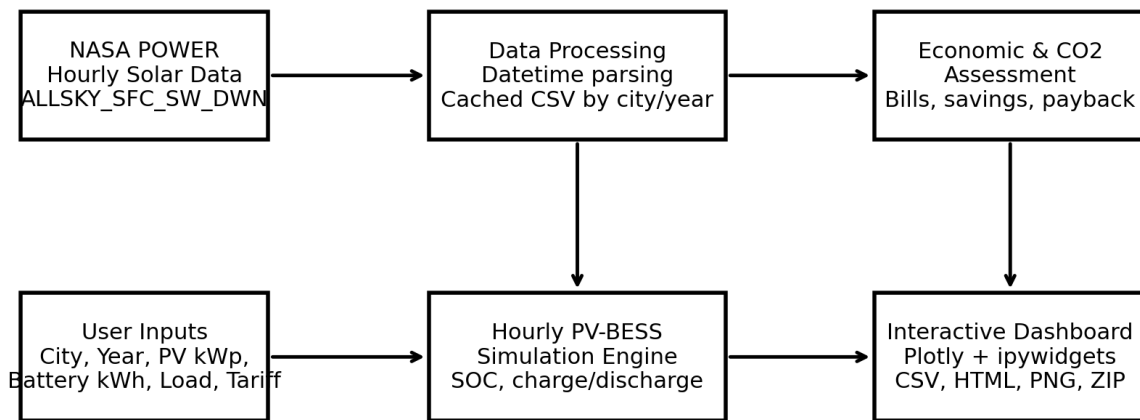


Fig. 1. System architecture of the Solardach Energie Dashboard.

Execution workflow from data acquisition to dashboard export

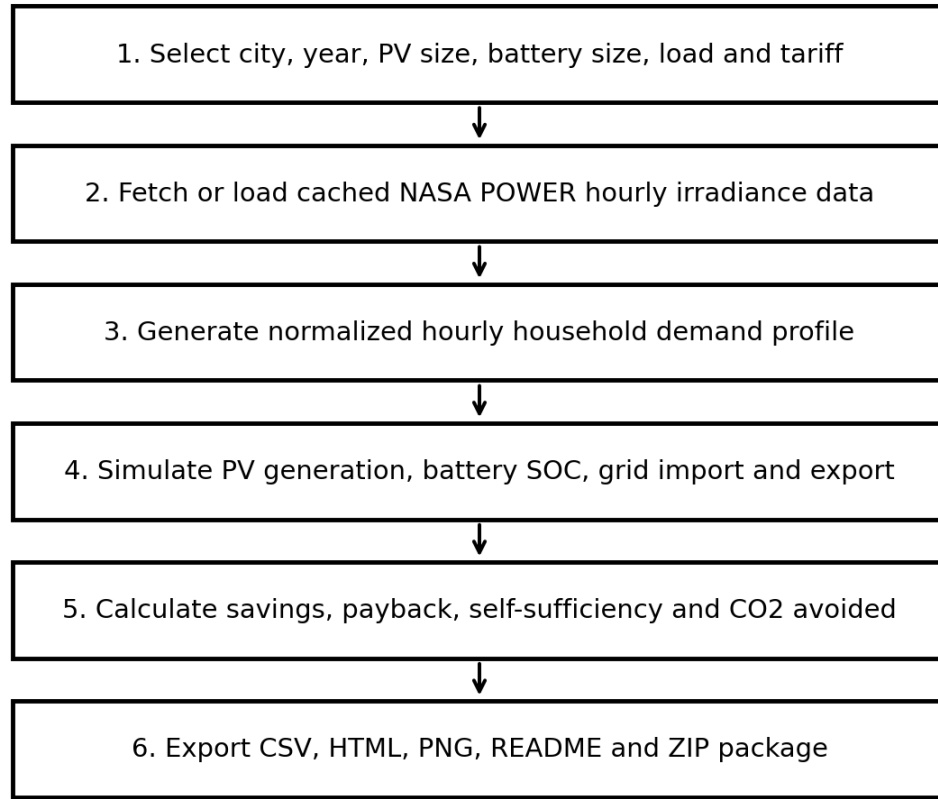


Fig. 2. Execution workflow from data acquisition to dashboard export.

V. SOFTWARE IMPLEMENTATION

The dashboard is implemented as a Google Colab notebook using Python. The principal libraries are requests for API access, pandas and NumPy for data processing, Plotly for interactive dashboard visualization, Matplotlib for saved static plots, ipywidgets for control sliders/dropdowns and zipfile/pathlib for export automation. The code contains modular functions for data acquisition, demand modelling, PV-BESS dispatch, economic calculation, tariff comparison, monthly summary generation, dashboard building and output saving.

Table III. Major functions and technical role in the submitted program.

Function	Technical Role
fetch_nasa_power_hourly	Downloads or loads cached hourly solar irradiance for a selected city and year.
create_hourly_demand_profile	Builds a normalized residential demand curve using hour, season and weekend factors.
simulate_pv_battery	Calculates hourly PV generation, self-consumption, battery SOC, grid import and grid export.
calculate_economics	Computes baseline bill, new grid bill, feed-in revenue, annual savings, payback and CO2 avoided.
compare_german_tariffs	Runs the economic calculation over all defined German tariff cases.
build_video_dashboard	Creates a 3x2 Plotly dashboard with monthly balance, hourly curves, SOC, tariff comparison and gauge.
save_outputs	Exports hourly CSV, monthly CSV, tariff CSV, dashboard HTML, PNG charts, README and ZIP package.

VI. DASHBOARD OUTPUTS

The dashboard is designed to produce both interactive and report-ready outputs. The interactive screen contains a title card with project name, author, selected city tier, annual PV generation, self-sufficiency, annual savings and annual CO2 avoided. The Plotly dashboard includes six panels: monthly energy balance, hourly solar-demand simulation on the best solar day, battery state of charge, tariff comparison, monthly grid import versus export and an energy independence gauge.

Table IV. Files generated by the Save outputs function.

Generated File	Purpose
hourly_pv_bess_simulation.csv	Complete hourly time-series including irradiance, demand, PV generation, battery charge, discharge, SOC, grid import and export.
monthly_energy_summary.csv	Aggregated monthly PV generation, demand, self-consumption, import and export.
german_tariff_comparison.csv	Baseline bill, post-PV-BESS grid bill, feed-in revenue, annual savings and payback for each tariff case.
video_dashboard.html	Interactive Plotly dashboard for presentation and screen recording.
video_chart_monthly_energy_balance.png	Static monthly chart for report or LinkedIn upload.
video_chart_tariff_comparison.png	Static tariff comparison chart.
video_chart_hourly_battery_simulation.png	Static best-solar-day hourly PV, demand, SOC and import chart.
README_video_project_summary.txt	Narration guide and file inventory for video presentation.

VII. RESULTS AND DISCUSSION

The submitted program runs the dashboard by default for Freiburg, year 2024, 7.0 kWp PV capacity, 8.0 kWh battery, 4200 kWh annual load, BDEW average household tariff and 0.82 PV performance ratio. Once executed in Colab with internet access, the simulation produces city-specific and year-specific results based on the fetched hourly irradiance data. The report does not fabricate numerical output values because the latest message provided code but did not attach the generated CSV/PNG output files from a completed run.

From a technical interpretation point of view, the dashboard is expected to show high daytime PV generation, strong summer production, lower winter production, battery charging during surplus periods and battery discharging during evening or low-solar periods. The tariff comparison panel demonstrates that economics are highly sensitive to the electricity import price. Higher import tariffs increase the value of self-consumed PV energy, while lower tariffs reduce annual savings. Battery size affects the self-sufficiency ratio by shifting surplus solar energy to later load periods, but payback depends on whether the additional avoided import is sufficient to justify the battery capital cost.

The model is especially useful for comparing how the same PV-BESS configuration behaves across city tiers. Southern and sunnier locations are likely to show stronger PV generation, while northern cities may require greater storage or careful sizing to achieve comparable self-sufficiency. However, final city ranking should be based on actual executed outputs, not assumptions.

VIII. NOVELTY AND RELEVANCE TO RENEWABLE ENERGY STUDIES

The project has academic relevance because it combines solar resource analytics, load modelling, storage dispatch, economics and visualization into one reproducible workflow. For a renewable energy or smart grid portfolio, this demonstrates practical skills in Python programming, energy data handling, PV-BESS modelling, tariff-aware decision support and dashboard communication. It is also suitable for extension into a more advanced MSc-level project by adding PVGIS validation, real smart meter load data, dynamic tariffs, battery degradation, net present value, LCOE and grid peak reduction metrics.

IX. LIMITATIONS

- 1) The demand profile is synthetic and should be replaced or calibrated with real household smart meter data for final research use.
- 2) The tariff values are embedded assumptions and should be updated from official or current market sources before publication or investment use.
- 3) Battery degradation is not explicitly modelled in the new dashboard code, although battery dispatch and SOC are simulated.
- 4) The current economic model uses simple payback and does not include NPV, inflation, inverter replacement, degradation, taxes or financing.
- 5) The model assumes a simplified PV conversion formula and does not include tilt, azimuth, shading, inverter clipping or module temperature losses beyond PR.

X. FUTURE SCOPE

- 1) Integrate PVGIS hourly data and compare it with NASA POWER irradiance.
- 2) Add net present value, internal rate of return, LCOE and sensitivity analysis.
- 3) Add battery degradation, replacement year and state-of-health modelling.
- 4) Add dynamic electricity tariffs and EV charging demand profiles.
- 5) Run batch optimization for all German cities and automatically recommend optimal PV and battery sizes.
- 6) Deploy the dashboard as a Streamlit web app or GitHub Pages interactive report.

XI. CONCLUSION

The Solardach Energie Dashboard is a strong applied renewable energy analytics project by Oihika Arpit. It converts hourly solar resource data and residential demand assumptions into an interactive PV-BESS performance and tariff analysis workflow. The code demonstrates a practical understanding of solar generation modelling, battery dispatch, electricity bill economics, self-sufficiency and dashboard-based technical communication. With additional validation using official solar datasets and real tariff updates, this project can be expanded into a high-quality academic portfolio project for renewable energy systems, smart grids and energy storage applications.

REFERENCES.

- [1] O. Arpit, "Solardach Energie Dashboard," submitted Python source code, 2026.
- [2] NASA POWER Project, hourly point solar radiation API endpoint used in the submitted implementation.
- [3] Submitted code tariff assumptions for German household import price, feed-in price and standing charge cases, 2026.
- [4] Python Software Foundation, Python programming language and standard libraries.
- [5] pandas, NumPy, Plotly, Matplotlib and ipywidgets libraries used for data processing, visualization and interactive controls.

APPENDIX A. CODE INVENTORY

The submitted code file defines the project name as Solardach Energie Dashboard and author name as Oihika Arpit. It installs required packages, enables Colab widget support, defines city and tariff dictionaries, contains functions for data retrieval, demand generation, PV-BESS simulation, economic calculation, dashboard creation and output export, and finally renders widget controls before running the default dashboard configuration.